

TECHNOLOGIES FOR CAPTURING MUSICAL GESTURES

STATE OF THE ART, APPLICATIONS, AND LIMITATIONS

ROMAIN MICHON

PHYSICAL INTERACTION DESIGN COURSE

UNIVERSITÉ JEAN MONNET, SAINT-ÉTIENNE



Caricature of Franz LISZT's performance movements (A. Göschl, 1873).

WHAT IS A MUSICAL GESTURE?

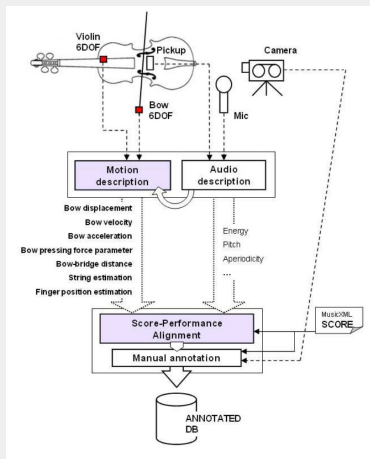


- **Instrumental gesture** (fingers, arms, posture).
- **Expressive gesture** (intention, phrasing, dynamics).
- **Whole-body gesture** (entire body, dance, conducting).

⇒ Physical, perceptual, and symbolic dimensions

REFSUM JENSENIUS, Alexander, *ACTION – SOUND Developing Methods and Tools to Study Music-Related Body Movement*, unpublished doctoral dissertation, University of Oslo, 2007.

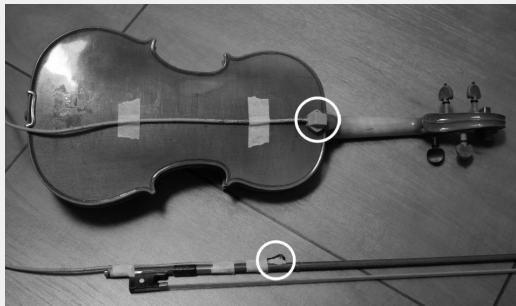
WHY CAPTURE MUSICAL GESTURES? → THE VIOLIN



Acquiring violin gesture data with Polhemus electromagnetic field (EMF) sensors providing 6DoF¹.

¹MAESTRE, Esteban; BONODA, Jordi; BLAAUW, Merlijn; PÉREZ, Alfonso; GUAUS, Enric, *Acquisition of Violin Instrumental Gestures Using a Commercial EMF Tracking Device: Proceedings of the International Computer Music Conference (ICMC-07), Copenhagen, 2007, Ann Arbor: ICMA, 2007.*

WHY CAPTURE MUSICAL GESTURES? → THE VIOLIN



Electromagnetic field (EMF) sensors placed on a violin and its bow.

- A physical model of a violin can sound highly synthetic when controlled with a MIDI keyboard
 - ▶ **Example**
- The same model sounds much better when controlled by gestural data
 - ▶ **Example**

MICHON, Romain; SMITH, Julius O., *Faust-STK: a Set of Linear and Nonlinear Physical Models for the Faust Programming Language*; Proceedings of the 14th International Conference on Digital Audio Effects (DAFx-11), Paris, 2011, Hamburg: DAFx, 2011.

OVERVIEW OF CAPTURE TECHNOLOGIES



The “famous” Terry Chest Drum, using electromyography sensors (EMG):
https://youtu.be/0hwpDnA0_V8

- Optical sensors (cameras, RGB, etc.).
- Sensors worn on the body or attached to the instrument (inertial, physiological, etc.).
- Hybrid and multimodal approaches.

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■ **Relative vs. absolute position**

- ▶ Is detecting acceleration enough, or is the exact 3D position required?

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- ▶ Sensor weight, wiring, etc.

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■ **Wireless vs. wired**

- ▶ An issue intrinsically linked to latency.

■ **Capture across a large stage**

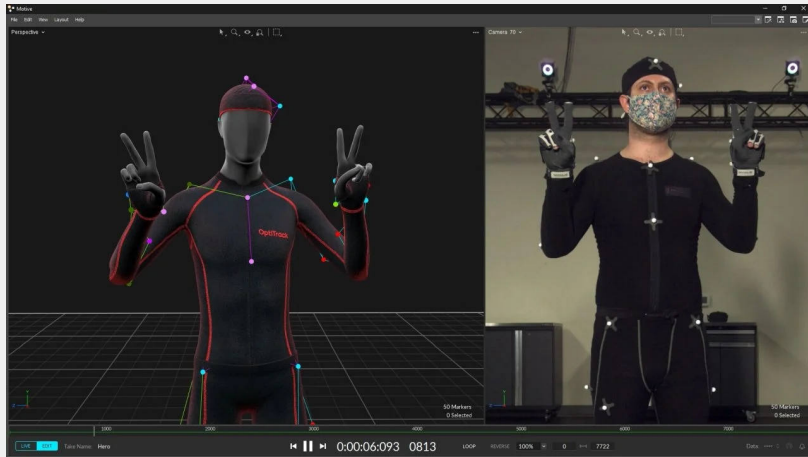
- ▶ What happens when the performer is highly mobile and moves extensively around the stage?

- **RGB cameras (standard cameras) and RGB-D cameras (with depth sensing)**
 - ▶ Manufacturers: Kinect, RealSense, etc.
 - ▶ Advantages: minimally intrusive, moderate cost
 - ▶ Disadvantages: sensitive to lighting and occlusion, limited finger-tracking accuracy, latency

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- **Multi-camera motion capture**
 - ▶ A network of synchronized cameras (often infrared) tracks reflective markers placed on the body or instrument, enabling 3D reconstruction through triangulation.
 - ▶ Manufacturers: Vicon, OptiTrack, Qualisys, etc.
 - ▶ Advantages: very high accuracy, the established standard
 - ▶ Disadvantages: controlled environment, cost, calibration time, intrusive

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- **Computer vision (markerless)**
 - ▶ Video analysis using computer-vision and AI algorithms: automatic detection of the body, hands, or fingers
 - ▶ Techniques: OpenPose, MediaPipe, etc.
 - ▶ Advantages: minimally intrusive, effective
 - ▶ Disadvantages: latency, sensitivity to viewing angles and occlusion, approximate 3D reconstruction

BEVILACQUA, Frédéric; SCHNELL, Norbert; ALAOUI, Sarah Fdili, “Gesture Capture: Paradigms and Applications in Music Performance and Composition,” *Emerging Bodies*, Paris: IRCAM, 2011.



OptiTrack system (Source: <https://optitrack.com/>).

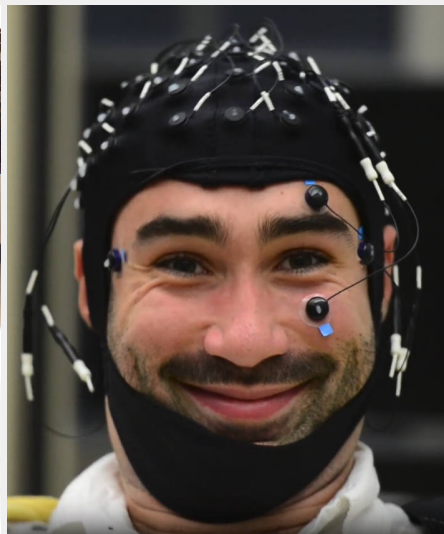
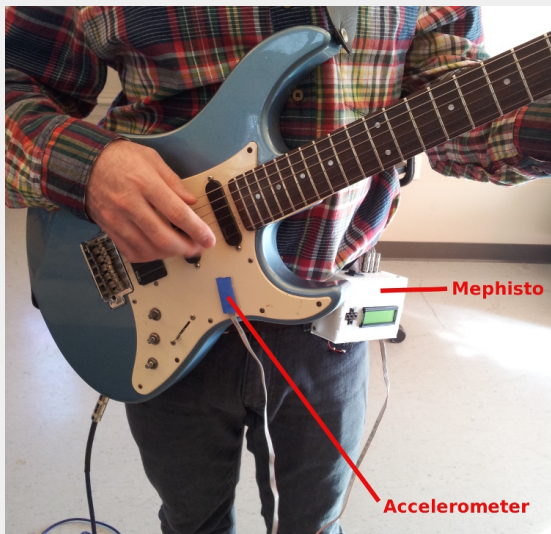
HandMate system (AI): <https://monlim.github.io/Handmate-MIDI/>

INERTIAL (IMU) AND PHYSIOLOGICAL SENSORS

- Accelerometer
 - ▶ Detects acceleration in 3D and is sensitive to Earth's gravity.
 - ▶ Easy to implement // intrusive // drift
- Gyroscope
 - ▶ Detects 3D orientation relative to Earth's gravity.
 - ▶ Easy to implement // works effectively with accelerometers // intrusive
- Magnetometer/EMF
 - ▶ Detects an absolute position in 3D
 - ▶ Very high accuracy (a widely used solution in music) // cumbersome to implement
- Electromyography (EMG)
 - ▶ Detects muscle contractions.
 - ▶ “Primary” data // noisy signal // highly intrusive
- Electroencephalography (EEG)
 - ▶ Detects electrical activity produced by the brain.
 - ▶ “Thought-based” control // extremely noisy signal that is complex to analyze // extremely intrusive

TANAKA, Atau. “Musical performance practice on sensor-based instruments,” *Trends in gestural control of music*, XIII (2000), pp. 389–405.

FIERO, David, *Creating Prototype Instruments to Make Music from Electrophysiological Signals*, unpublished doctoral dissertation, Paris 8 University, 2025.



Accelerometer attached to a guitar (left) and an EEG headset used in a study of the singing voice (right).

THE ROLE OF AI IN MUSICAL GESTURE CAPTURE

- Greater effectiveness of systems based on ordinary cameras (a wide range of gestures can now be detected using only a webcam), as well as other systems.
- Greater capacity for analysis and extrapolation: gesture recognition, style classification, learning relationships between gesture and expressivity, etc.
- Latency remains an issue for real-time applications.
- Techniques are evolving very rapidly, and much of the research on musical gesture capture is moving in this direction.
- Toward methods based exclusively on sound?

→ Toward adaptive and expressive systems

FIEBRINK, Rebecca; COOK, Perry R.; TRUEMAN, Dan. *Human Model Evaluation in Interactive Supervised Learning: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*, New York, 2011. Association for Computing Machinery: New York, 2011.

PELINSKI, Teresa; SHEPARDSON, Victor; SYMONS, Steve; CAPSE, Franco Santiago; TEMPRANO, Adan L. Benito; ARMITAGE, Jack; KIEFER, Chris; FIEBRINK, Rebecca; MAGNUSSON, Thor; MCPHERSON, Andrew, *Embedded AI for NIME: Challenges and Opportunities: Proceedings of the International Conference on New Interfaces for Musical Expression (NIME 2022)*, Auckland, New Zealand, 2022, NIME: Auckland, 2022.